

Electronic Supplement ST-01: Included Studies and Eligible Report Lineage for the O-ISAC Survey

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PURPOSE AND COUNTING BOUNDARY

This supplement identifies and characterizes every study included in the systematic survey. The review unit is the unique study: 227 eligible reports were mapped to 206 studies, so companion reports do not appear as independent technical demonstrations. Each study row cites its designated primary report and every eligible companion report. ST-01B exposes the exact evidence, metric, and tradeoff provenance of the 21 companions. A historical `pending_human` token triggers exclusion only when no resolved or approved adjudication is recorded; report-specific conflict guardrails remain in force after adjudication.

Electronic Table ST-01A: Included-study inventory for the 206 unique studies
Each study row cites its primary report and every eligible companion report; companions remain within the same study.

Study ID	Report citations	Title	Modality	Integration mechanisms	Validation	Reports
STC-ACO-OFDM-LFM-LIDAR-2025	[1]	Integrated Design of OFDM-LFM Lidar Communication in IoV Environment	FSO	shared_waveform; shared_resource_allocation	simulation; experiment	laboratory_1
STC-DP-IMDD-T HZ-RANGING-2025-2026	[2], [3]	310Gbps Transmission and Centimeter-Level Ranging Photonic THz ISAC System	photonic_THz	shared_hardware; shared_resource_allocation	analytical; experiment; prototype_testbed	laboratory_2
STC-DSCM-ANN-DT-ECL-2025-2026	[4], [5]	Machine-learning-enhanced phase-based vibration sensing in coherent DSCM systems under commercial ECLs	fiber	shared_waveform; shared_resource_allocation	simulation; dataset_based; laboratory_experiment	2
STC-EADO-OFDM-RANGING-2024-2025	[6], [7]	Optical Wireless Integrated Sensing and Communication Based on EADO-OFDM: A Flexible Resource Allocation Perspective	FSO	shared_waveform; shared_resource_allocation	analytical; simulation	2
STC-ENDOGENOUS-TS-FRFT-DAS-2025	[8], [9]	Endogenous integration of communication and interference fading free sensing using telecom pilots via joint polarization-fraction domain multiplexing	fiber	shared_waveform; shared_link_or_channel; shared_resource_allocation	laboratory_experiment	2
STC-FRDM-POLARIZATION-SENSING-2025-2026	[10], [11]	Integrated Communication and In-Band Spectrum Polarization-Based Sensing via Fraction-Division Non-Orthogonal Multiple Access	fiber	shared_waveform; shared_hardware; shared_link_or_channel; shared_resource_allocation	laboratory_experiment	2
STC-ICP-RL-HLLB-3650KM-2024-2025	[12]	Simultaneous Transmission and Sensing Emulation Using Interconnected Counter-Propagating Recirculating Loops	fiber	shared_link_or_channel	simulation; experiment	laboratory_1

Electronic Table ST-01A (continued): studies 8–14 of 206
Each study row cites its primary report and every eligible companion report; companions remain within the same study.

Study ID	Report citations	Title	Modality	Integration mechanisms		Validation	Reports
STC-III-NITRIDE-MOTION-VLC-2023	[13], [14]	Integrated Sensing and Communication Chip Based on III-Nitride for Motion Detection	VLC_LiFi	shared_hardware; channel	shared_link_or_	laboratory_experiment; prototype_testbed; mixed	2
STC-LED-OISAC-2024-2025	[15]	Optical Integrated Sensing and Communication With Light-Emitting Diode	VLC_LiFi	shared_resource_allocation		analytical; simulation	1
STC-MMW-OFD-M-PENG-2023-2024	[16], [17]	Photonics-assisted integrated sensing and communication with ranging resolution improvement by multiple signal classification	photonic_THz	shared_waveform; shared_resource_allocation	shared_hardware;	simulation; laboratory_experiment; mixed	2
STC-MULTICHANNEL-LFM-QAM-THZ-2024-2026	[18], [19]	Multi-Channel Photonic THz-ISAC System Based on Integrated LFM-QAM Waveform	photonic_THz	shared_hardware; allocation; optimization	shared_resource_joint_design_or_	analytical; simulation; laboratory_experiment; prototype_testbed	2
STC-OCDM-THZ-OVER-FIBER-2024-2025	[20], [21]	THz-Over-Fiber System With Orthogonal Chirp Division Multiplexing for Integrated Sensing and Communication	photonic_THz	shared_waveform; allocation	shared_resource_	analytical; simulation; laboratory_experiment	2
STC-ODDM-PHOTONIC-MMW-2024-2025	[22], [23]	Photonics-Assisted mmWave Integrated Sensing and Communication System Enabled by ODDM Waveform	photonic_THz	shared_waveform; shared_resource_allocation	shared_hardware;	analytical; simulation; laboratory_experiment; prototype_testbed	2
STC-OPA-OWISAC-BEAMFORMING-2024-2025	[24], [25]	Optical Wireless Integrated Sensing and Communication Based on Optical Phased Array: Performance Metric and Optimal Beamforming	FSO	joint_design_or_optimization		analytical; numerical_analysis; simulation	2

Electronic Table ST-01A (continued): studies 15–21 of 206
Each study row cites its primary report and every eligible companion report; companions remain within the same study.

Study ID	Report citations	Title	Modality	Integration mechanisms	Validation	Reports
STC-REALTIME-BIDIR-DMT-THZ-2024-2026	[26], [27], [28]	Real-Time Integrated Sensing with 5.08 mm Range Resolution and 15.5 Gbps Bidirectional Wireless Communication Over a 270-Meter Outdoor Photonic-Assisted Terahertz Link	photonic_THz	shared_waveform; shared_hardware	analytical; simulation; prototype_testbed; field_experiment	3
STC-SELF-HOMODYNE-MCF-OFDR-2024-2025	[29], [30]	Integrated Sensing and Communication in Self-Homodyne MCF Transmission Systems With Linear Frequency Modulation	fiber	shared_optical_carrier	analytical; laboratory_experiment	2
STC-SEVENCOR E-COMB-SH-DAS-2024-2025	[31], [32]	Integrating High-Capacity Self-Homodyne Transmission and High-Sensitivity Dual-Pulse ϕ -OTDR with an EO Comb over a 7-Core Fiber	fiber	shared_hardware; shared_optical_carrier; shared_link_or_channel; shared_resource_allocation; joint_design_or_optimization	analytical; laboratory_experiment; prototype_testbed	2
STC-SF-LFM-OFDM-THZ-2025-2026	[33], [34]	Photonics-Aided THz Integrated Sensing and Communication System Based on a Time-Frequency Efficient SF-LFM-OFDM Waveform	photonic_THz	shared_waveform; shared_optical_carrier; shared_resource_allocation	analytical; simulation; laboratory_experiment	2
STC-SINGLETON-SCR-00001	[35]	High-Speed Flexible Near-Infrared Organic Photodetectors for Self-Powered Optical Integrated Sensing and Communications	VLC_LiFi	shared_hardware; shared_application_scenario	laboratory_experiment; prototype_testbed	1
STC-SINGLETON-SCR-00002	[36]	Integrated Optical Sensing and Communication (IOSAC) System Based on Hybrid-Waveguide Structures	fiber	shared_hardware; shared_optical_carrier; shared_link_or_channel; joint_design_or_optimization	numerical_analysis; laboratory_experiment; prototype_testbed	1
STC-SINGLETON-SCR-00007	[37]	OFDM-based optical integrated sensing and communication framework using embedded pilot subcarriers	fiber	shared_waveform; shared_link_or_channel; shared_resource_allocation	analytical; simulation	1

Electronic Table ST-01A (continued): studies 22–28 of 206
Each study row cites its primary report and every eligible companion report; companions remain within the same study.

Study ID	Report citations	Title	Modality	Integration mechanisms	Validation	Reports
STC-SINGLETON-SCR-00008	[38]	Enhanced single-fiber bi-directional FT-DVS using a 600-kHz linewidth DBR laser with LPN compensation for O-ISAC [Invited]	fiber	shared_hardware; shared_link_or_channel; shared_application_scenario	mixed	1
STC-SINGLETON-SCR-00009	[39]	Optical integrated sensing and communication (O-ISAC): A unified framework for 6G and AI-Powered intelligent networks	hybrid_optical	shared_waveform; shared_hardware; shared_resource_allocation; joint_design_or_optimization	numerical_analysis; simulation	1
STC-SINGLETON-SCR-00011	[40]	Retroreflective optical ISAC using OFDM: channel modeling and performance analysis	FSO	shared_waveform; shared_hardware; shared_optical_carrier; shared_link_or_channel	analytical; simulation; laboratory_experiment	1
STC-SINGLETON-SCR-00012	[41]	Single-Source optical integrated sensing and communication (O-ISAC) with sequential target recognition	FSO	shared_optical_carrier; link_or_channel; shared_resource_allocation; mixed	laboratory_experiment; prototype_testbed	1
STC-SINGLETON-SCR-00013	[42]	On the coexistence of distributed fiber optic sensing and IM/DD transmission via digital subcarrier multiplexing	fiber	shared_waveform; shared_hardware; shared_optical_carrier; shared_link_or_channel; joint_design_or_optimization	laboratory_experiment; prototype_testbed	1
STC-SINGLETON-SCR-00015	[43]	Integrated optical covert sensing and communication	fiber	shared_waveform; shared_hardware; shared_optical_carrier; shared_link_or_channel	analytical; laboratory_experiment; prototype_testbed	1
STC-SINGLETON-SCR-00036	[44]	Millimeter-Wave Integrated Sensing and Communication System Based on a Photonics-Assisted Phased Array	photonic_THz	shared_waveform; shared_hardware; shared_resource_allocation	laboratory_experiment	1

Electronic Table ST-01A (continued): studies 29–35 of 206
Each study row cites its primary report and every eligible companion report; companions remain within the same study.

Study ID	Report citations	Title	Modality	Integration mechanisms	Validation	Reports
STC-SINGLETON-SCR-00038	[45]	Native and Reconfigurable Distributed Acoustic Sensing Integrated with Single-carrier 64QAM Transmission Signal	fiber	shared_waveform; shared_link_or_channel; shared_resource_allocation	laboratory_experiment	1
STC-SINGLETON-SCR-00052	[46]	Cc-Otdr Sequence Shaping Enabling Joint Co-Directional Sensing and Communication	fiber	shared_link_or_channel; shared_resource_allocation	laboratory_experiment	1
STC-SINGLETON-SCR-00056	[47]	Joint break-point localization sensing based on OFDR and non-disruptive communication fiber ISAC system	fiber	shared_waveform; shared_link_or_channel; shared_resource_allocation	analytical; simulation; laboratory_experiment; mixed	1
STC-SINGLETON-SCR-00057	[48]	Pre-Compensation of LFM Sensing Probe via Integrated Sensing and Communication over Fiber System	fiber	shared_link_or_channel; shared_resource_allocation	laboratory_experiment	1
STC-SINGLETON-SCR-00086	[49]	Photonic THz Sensing-assisted Multipath Channel Estimation for ISAC	photonic_THz	shared_waveform; shared_resource_allocation	laboratory_experiment	1
STC-SINGLETON-SCR-00091	[50]	Communication and Sensing via Hermite-Gaussian Time-Domain Mode-Division-Multiple-Access	hybrid_optical	shared_waveform; shared_resource_allocation; joint_design_or_optimization	numerical_analysis; simulation	1
STC-SINGLETON-SCR-00110	[51]	Integrated Optical Camera Communication and Scene Sensing Based on Generative Adversarial Networks	VLC_LiFi	shared_waveform; shared_optical_carrier; joint_design_or_optimization; shared_application_scenario	dataset_based; laboratory_experiment; mixed	1

Electronic Table ST-01A (continued): studies 36–42 of 206
Each study row cites its primary report and every eligible companion report; companions remain within the same study.

Study ID	Report citations	Title	Modality	Integration mechanisms	Validation	Reports
STC-SINGLETON-SCR-00123	[52]	Glass fiber-based integrated sensing and communication system for vehicular applications	fiber	shared_link_or_channel; application_scenario	shared_laboratory_experiment; prototype_testbed	1
STC-SINGLETON-SCR-00151	[53]	Integrated Communication and Enhanced Forward Phase-Based Sensing Based on Frequency-Domain Pilot Tones in DSCM Systems Using 100 kHz ECLs	fiber	shared_waveform; shared_link_or_channel; resource_allocation; joint_design_or_optimization	shared_hardware; shared_experiment; mixed	1
STC-SINGLETON-SCR-00160	[54]	Photonic Generation of Joint Amplitude-Frequency Modulated Waveform for THz Integrated Sensing and Communication	photonic_THz	shared_waveform; shared_optical_carrier; or_optimization	shared_hardware; joint_design_prototype_testbed	1
STC-SINGLETON-SCR-00177	[55]	Integrated photonic ultrawideband real-time spectrum sensing for 6G wireless networks	photonic_THz	shared_resource_allocation; joint_design_or_optimization; application_scenario	simulation; laboratory_experiment; prototype_testbed; mixed	1
STC-SINGLETON-SCR-00185	[56]	Integrated Terahertz FMCW Radar and FSK Communication Enabled by High-Speed Wavelength Tunable Lasers	photonic_THz	shared_hardware; allocation	shared_resource_laboratory_experiment; prototype_testbed	1
STC-SINGLETON-SCR-00189	[57]	Sensing μm -scale vibrations in the Hz range within a THz communication system	photonic_THz	shared_waveform; shared_optical_carrier; link_or_channel; joint_design_or_optimization	shared_hardware; shared_analytical; dataset_based; laboratory_experiment; prototype_testbed; mixed	1
STC-SINGLETON-SCR-00196	[58]	Adaptive visible light integrated sensing and communication for cable-free virtual reality	VLC_LiFi	shared_waveform; shared_optical_carrier; link_or_channel; joint_design_or_optimization	shared_hardware; shared_laboratory_experiment; prototype_testbed	1

Electronic Table ST-01A (continued): studies 43–49 of 206
Each study row cites its primary report and every eligible companion report; companions remain within the same study.

Study ID	Report citations	Title	Modality	Integration mechanisms	Validation	Reports
STC-SINGLETON-SCR-00210	[59]	Software-Defined Radio Experimental Validation of an OTFS-Based ISAC for Velocity Estimation in an ARoF Setup	hybrid_optical	shared_waveform; shared_link_or_channel; joint_design_or_optimization	simulation; experiment; mixed	laboratory_prototype_testbed; 1
STC-SINGLETON-SCR-00218	[60]	Parallel Path Effect-Free Remote Monitoring and Sensing in Live XGS-PON Using Legacy Optical Time Domain Reflectometry	fiber	shared_hardware; shared_link_or_channel; shared_application_scenario	analytical; numerical_analysis; laboratory_experiment; prototype_testbed	1
STC-SINGLETON-SCR-00233	[61]	Bi-LSTM-Transformer hybrid network for V2V optical wireless ISAC systems	FSO	shared_waveform; shared_optical_carrier; shared_link_or_channel; joint_design_or_optimization	simulation	1
STC-SINGLETON-SCR-00238	[62]	Multi-service integrated sensing and communication system for industrial internet-of-things	fiber	shared_waveform; shared_optical_carrier; shared_link_or_channel; shared_application_scenario	laboratory_experiment; prototype_testbed	1
STC-SINGLETON-SCR-00256	[63]	OCDM Integrated LiDAR and Communication System for UAVs	FSO	shared_waveform; shared_optical_carrier; shared_link_or_channel; joint_design_or_optimization	analytical; simulation	1
STC-SINGLETON-SCR-00268	[64]	Full dimensional control of structured microwaves based on microcombs	other_optical	shared_hardware; shared_application_scenario	numerical_analysis; laboratory_experiment; prototype_testbed	1
STC-SINGLETON-SCR-00273	[65]	Optical Camera-Based Integrated Sensing and Communication for V2X Applications: Model and Optimization	VLC_LiFi	shared_hardware; shared_resource_allocation; joint_design_or_optimization; shared_application_scenario	analytical; numerical_analysis; laboratory_experiment; prototype_testbed	1

Electronic Table ST-01A (continued): studies 50–56 of 206
Each study row cites its primary report and every eligible companion report; companions remain within the same study.

Study ID	Report citations	Title	Modality	Integration mechanisms	Validation	Reports
STC-SINGLETON-SCR-00274	[66]	A Unified OFDM-ISAC Signal Generation Architecture in W-Band via Photonics-Aided Frequency Multiplication and Phase Noise Mitigation	photonic_THz	shared_waveform; shared_resource_allocation; shared_design_or_optimization; shared_hardware; joint_simulation	analytical; numerical_analysis; simulation	1
STC-SINGLETON-SCR-00277	[67]	Towards last-mile connectivity in 6G-IoT: An integrated MIMO-FSO communication system and FBG sensors under atmospheric attenuations, strain and temperature effects	hybrid_optical	shared_hardware; shared_link_or_channel; joint_design_or_optimization; shared_application_scenario	analytical; numerical_analysis; simulation	1
STC-SINGLETON-SCR-00278	[68]	Demonstration of High Sensitivity Detection and High Precision Localization of Vibrations in Bidirectional Coherent Transmission Systems Enabled by Carrier Laser Phase Noise Retrieval	fiber	shared_waveform; shared_optical_carrier; shared_link_or_channel; joint_design_or_optimization	laboratory_experiment; prototype_testbed	1
STC-SINGLETON-SCR-00294	[69]	Optoelectronic integrated sensing and communication system based on InGaN/GaN MQW for wireless detection of air pressure	VLC_LiFi	shared_hardware; joint_design_or_optimization; shared_application_scenario	laboratory_experiment; prototype_testbed	1
STC-SINGLETON-SCR-00312	[70]	Photonics-Aided THz ISAC System Based on A Spectral Overlapping Multi-Carrier 16QAM-LFM Waveform	photonic_THz	shared_waveform; shared_optical_carrier; shared_resource_allocation; joint_design_or_optimization	laboratory_experiment; prototype_testbed	1
STC-SINGLETON-SCR-00344	[71]	Co-wavelength-channel integration of ultra-low-frequency distributed acoustic sensing and high-capacity communication	fiber	shared_hardware; shared_optical_carrier; shared_link_or_channel; shared_resource_allocation; joint_design_or_optimization	simulation; laboratory_experiment; prototype_testbed	1
STC-SINGLETON-SCR-00356	[72]	Exploitation of FMF capabilities for joint communications and sensing	fiber	shared_waveform; shared_optical_carrier; shared_hardware; shared_link_or_channel	laboratory_experiment; prototype_testbed	1

Electronic Table ST-01A (continued): studies 57–63 of 206
Each study row cites its primary report and every eligible companion report; companions remain within the same study.

Study ID	Report citations	Title	Modality	Integration mechanisms		Validation		Reports
STC-SINGLETON-SCR-00361	[73]	Photonic-aided Doppler-robust ISAC system based on index modulation and chirp multiplexing	photonic_THz	shared_waveform; shared_optical_carrier; resource_allocation; optimization	shared_hardware; shared_resource_allocation; joint_design_or_optimization	analytical; laboratory_experiment; prototype_testbed; mixed	simulation;	1
STC-SINGLETON-SCR-00362	[74]	Frequency-comb-steered ultrawideband quasi-true-time-delay beamformer for integrated sensing and communication	photonic_THz	shared_waveform; shared_optical_carrier; resource_allocation; optimization	shared_hardware; shared_resource_allocation; joint_design_or_optimization	analytical; laboratory_experiment; prototype_testbed; mixed	simulation;	1
STC-SINGLETON-SCR-00366	[75]	Integrated ultra-large dynamic vibration sensing with fronthaul analog radio-over-fiber transmission	hybrid_optical	shared_waveform; shared_optical_carrier; link_or_channel; shared_resource_allocation; optimization	shared_hardware; shared_resource_allocation; joint_design_or_optimization	simulation; experiment; mixed	laboratory_prototype_testbed;	1
STC-SINGLETON-SCR-00367	[76]	Common bias-digital subcarrier multiplexing scheme for integrated sensing and communication in short-reach optical transmission systems	fiber	shared_waveform; shared_optical_carrier; link_or_channel; shared_resource_allocation; optimization	shared_hardware; shared_resource_allocation; joint_design_or_optimization	analytical; experiment; prototype_testbed	laboratory_prototype_testbed	1
STC-SINGLETON-SCR-00369	[77]	Simultaneous polarization demultiplexing and vibration sensing via single pilot tone in DSCM-ISAC systems	fiber	shared_waveform; shared_optical_carrier; link_or_channel; shared_resource_allocation; optimization	shared_hardware; shared_resource_allocation; joint_design_or_optimization	analytical; experiment; prototype_testbed	laboratory_prototype_testbed	1
STC-SINGLETON-SCR-00376	[78]	An integrated sensing and communication solar skin for health monitoring and human-machine interaction	VLC_LiFi	shared_hardware; shared_resource_allocation; optimization; scenario	shared_resource_allocation; shared_application_scenario	dataset_based; experiment; mixed	laboratory_prototype_testbed;	1
STC-SINGLETON-SCR-00379	[79]	Enhanced Room Coverage With Photon-Level NLOS Ultraviolet Optical Wireless Communication	other_optical	shared_waveform; shared_optical_carrier; or_channel; shared_application_scenario	shared_hardware; shared_link_or_channel; shared_application_scenario	laboratory_experiment; prototype_testbed		1

Electronic Table ST-01A (continued): studies 64–70 of 206
Each study row cites its primary report and every eligible companion report; companions remain within the same study.

Study ID	Report citations	Title	Modality	Integration mechanisms		Validation		Reports
STC-SINGLETON-SCR-00387	[80]	Demonstration of Ranging-Based Phase Encryption for Enhanced Free-Space Optical Security	FSO	shared_waveform; shared_optical_carrier; link_or_channel; optimization;	shared_hardware; shared_link_or_channel; joint_design_or_shared_application_scenario	analytical; laboratory_experiment; prototype_testbed	simulation;	1
STC-SINGLETON-SCR-00396	[81]	Integrated sensing and communication system based on a photonic integrated broadband multi-beamformer	photonic_THz	shared_hardware; shared_link_or_channel; shared_resource_allocation; shared_application_scenario	shared_hardware; shared_application_scenario	analytical; laboratory_experiment; prototype_testbed	simulation;	1
STC-SINGLETON-SCR-00406	[82]	Microcomb-synchronized optoelectronics	photonic_THz	shared_hardware; shared_application_scenario	shared_hardware; shared_application_scenario	analytical; laboratory_experiment; prototype_testbed	laboratory_experiment; prototype_testbed	1
STC-SINGLETON-SCR-00407	[83]	Analysis of Visible Light-Based ISAC Channel Characteristics in a V2V Scenario	VLC_LiFi	shared_optical_carrier; link_or_channel; shared_application_scenario	shared_link_or_channel; shared_application_scenario	analytical; numerical_analysis; simulation		1
STC-SINGLETON-SCR-00418	[84]	W-band photonics-aided OFDM system integrating sensing and communication with phase noise suppression scheme	photonic_THz	shared_waveform; shared_optical_carrier; link_or_channel; allocation; optimization	shared_hardware; shared_link_or_channel; shared_resource_allocation; joint_design_or_optimization	analytical; laboratory_experiment; prototype_testbed; mixed	simulation;	1
STC-SINGLETON-SCR-00432	[85]	Ultrasonic phase extraction method for co-cable identification in coherent optical transmission systems	fiber	shared_waveform; shared_optical_carrier; link_or_channel; joint_design_or_optimization	shared_hardware; shared_link_or_channel; joint_design_or_optimization	analytical; laboratory_experiment; prototype_testbed; mixed	simulation;	1
STC-SINGLETON-SCR-00436	[86]	Optical Transport Network Optimization Supporting Integrated Sensing and Communication Services	hybrid_optical	shared_waveform; shared_link_or_channel; resource_allocation; or_optimization; shared_application_scenario	shared_hardware; shared_link_or_channel; shared_resource_allocation; joint_design_or_shared_application_scenario	simulation; laboratory_experiment; prototype_testbed; mixed	laboratory_experiment; prototype_testbed;	1

Electronic Table ST-01A (continued): studies 71–77 of 206
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Study ID	Report citations	Title	Modality	Integration mechanisms	Validation	Reports
STC-SINGLETON-SCR-00451	[87]	Wearable Sensors Based on Miniaturized High-Performance Hybrid Nanogenerator for Medical Health Monitoring	VLC_LiFi	shared_hardware; optimization; shared_application_scenario	laboratory_experiment; prototype_testbed	1
STC-SINGLETON-SCR-00456	[88]	An Adaptive Control System for Satellite-Ground Laser Communication Based on Integrated Sensing and Communication	FSO	shared_waveform; shared_optical_carrier; shared_link_or_channel; joint_design_or_optimization; shared_application_scenario	analytical; simulation	1
STC-SINGLETON-SCR-00460	[89]	Integrated high precision time-of-flight depth sensing and optical communication based on Hamiltonian coding	VLC_LiFi	shared_waveform; shared_optical_carrier; shared_link_or_channel; shared_resource_allocation; joint_design_or_optimization	analytical; simulation; laboratory_experiment; prototype_testbed; mixed	1
STC-SINGLETON-SCR-00473	[90]	High-performance Vibration Sensing Based on Subband Phase-Shift Transform in Integrated Sensing and Communication System	fiber	shared_hardware; shared_link_or_channel; shared_resource_allocation; joint_design_or_optimization	analytical; laboratory_experiment; prototype_testbed; mixed	1
STC-SINGLETON-SCR-00488	[91]	6-DoF Location-and-Pose Estimation Toward Integrated Visible Light Communication and Sensing: Algorithm Design and Performance Limits	VLC_LiFi	shared_waveform; shared_link_or_channel; joint_design_or_optimization; shared_application_scenario	analytical; simulation	1
STC-SINGLETON-SCR-00492	[92]	Millimeter-wave over fiber integrated sensing and communication system using self-coherent OFDM	photonic_THz	shared_waveform; shared_optical_carrier; shared_link_or_channel; shared_resource_allocation; joint_design_or_optimization; shared_application_scenario	analytical; simulation; laboratory_experiment; prototype_testbed	1
STC-SINGLETON-SCR-00496	[93]	First Demonstration of FTTR-based RFID Network for Integrated Sensing and Communication	fiber	shared_hardware; shared_link_or_channel; shared_resource_allocation; shared_application_scenario	simulation; laboratory_experiment; prototype_testbed; field_experiment	1

Electronic Table ST-01A (continued): studies 78–84 of 206
Each study row cites its primary report and every eligible companion report; companions remain within the same study.

Study ID	Report citations	Title	Modality	Integration mechanisms	Validation	Reports
STC-SINGLETON-SCR-00502	[94]	Demonstration of radar-aided flexible communication in a photonics-based W-band distributed integrated sensing and communication system for 6G	photonic_THz	shared_hardware; shared_optical_carrier; shared_link_or_channel; shared_resource_allocation; joint_design_or_optimization; shared_application_scenario	analytical; laboratory_experiment; prototype_testbed	1
STC-SINGLETON-SCR-00506	[95]	Catch Me If You Can: Laser Tethering with Highly Mobile Targets	FSO	shared_hardware; shared_link_or_channel; joint_design_or_optimization; shared_application_scenario	laboratory_experiment; prototype_testbed; mixed	1
STC-SINGLETON-SCR-00527	[96]	Joint Visible Light Sensing and Communication Using m-CAP Modulation	VLC_LiFi	shared_waveform; shared_hardware; shared_optical_carrier; shared_link_or_channel; shared_resource_allocation; joint_design_or_optimization; shared_application_scenario	analytical; simulation; mixed	1
STC-SINGLETON-SCR-00557	[97]	Visible Light Communication-Enabled Simultaneous Position and Orientation Detection for Harnessing Multipath Interference and Random Fading	VLC_LiFi	shared_waveform; shared_hardware; shared_optical_carrier; shared_link_or_channel; joint_design_or_optimization; shared_application_scenario	analytical; simulation; mixed	1
STC-SINGLETON-SCR-00576	[98]	Remote Phase-Shift LiDAR With Communication	FSO	shared_waveform; shared_hardware; shared_optical_carrier; shared_link_or_channel; shared_resource_allocation; joint_design_or_optimization; shared_application_scenario	analytical; simulation; laboratory_experiment; mixed	1
STC-SINGLETON-SCR-00589	[99]	Deep Integration Between Polarimetric Forward-Transmission Fiber-Optic Communication and Distributed Sensing Systems	fiber	shared_waveform; shared_hardware; shared_link_or_channel	analytical; laboratory_experiment; prototype_testbed; mixed	1
STC-SINGLETON-SCR-00594	[100]	Wide-Steering Integrated Sensing and Communication OPA-Based Lidar Design with Joint Waveform and Thinned Antenna	FSO	shared_waveform; shared_hardware; joint_design_or_optimization	analytical; simulation; mixed	1

Electronic Table ST-01A (continued): studies 85–91 of 206
Each study row cites its primary report and every eligible companion report; companions remain within the same study.

Study ID	Report citations	Title	Modality	Integration mechanisms	Validation	Reports
STC-SINGLETON-SCR-00602	[101]	Photonics-assisted self-interference cancellation for in-band full-duplex integrated sensing and communication transceiver	photonic_THz	shared_waveform; shared_resource_allocation; joint_design_or_optimization	analytical; experiment; mixed	laboratory_prototype_testbed; 1
STC-SINGLETON-SCR-00607	[102]	Integrated Distributed Sensing and Quantum Communication Networks	fiber	shared_waveform; shared_link_or_channel; shared_resource_allocation	analytical; experiment; mixed	laboratory_prototype_testbed; 1
STC-SINGLETON-SCR-00626	[103]	Learned Digital Back-Propagation for an optical-domain waveform integrated with communication and sensing	fiber	shared_waveform; shared_optical_carrier; shared_link_or_channel; joint_design_or_optimization	simulation	1
STC-SINGLETON-SCR-00631	[104]	Integration of endogenous sensing and communication based on photonics-enabled cellular base station networks with a spectrum fusion algorithm	fiber	shared_waveform; shared_resource_allocation; joint_design_or_optimization; shared_application_scenario	analytical; simulation; laboratory_experiment; prototype_testbed; field_experiment	1
STC-SINGLETON-SCR-00647	[105]	Integrated sensing and communication in an optical fibre	fiber	shared_waveform; shared_optical_carrier; shared_link_or_channel; shared_resource_allocation; joint_design_or_optimization	analytical; experiment; laboratory_experiment; prototype_testbed	1
STC-SINGLETON-SCR-00665	[106]	Flexible Vibration Sensing Using Kalman Smoother Integrated in ECL-Based Communication Systems	fiber	shared_hardware; shared_optical_carrier; shared_link_or_channel; shared_application_scenario	simulation; experiment	laboratory_ 1
STC-SINGLETON-SCR-00679	[107]	A novel passive bi-directional audio over fiber transmission system with 20 Km	fiber	shared_hardware; shared_link_or_channel; shared_resource_allocation; shared_application_scenario	analytical; experiment; mixed	laboratory_ 1

Electronic Table ST-01A (continued): studies 92–98 of 206
Each study row cites its primary report and every eligible companion report; companions remain within the same study.

Study ID	Report citations	Title	Modality	Integration mechanisms	Validation	Reports
STC-SINGLETON-SCR-00680	[108]	Deep Integration of Fiber-Optic Communication and Sensing Systems Using Forward-Transmission Distributed Vibration Sensing and on–off Keying	fiber	shared_waveform; shared_optical_carrier; link_or_channel; joint_design_or_optimization; shared_application_scenario	shared_hardware; shared_experiment; analytical; mixed	laboratory_1
STC-SINGLETON-SCR-00719	[109]	Development of an Integrated Communication and Sensing System Using Spread Spectrum and Photonics Technologies	fiber	shared_waveform; shared_optical_carrier; link_or_channel; shared_resource_allocation; shared_application_scenario	shared_hardware; shared_experiment; analytical; simulation; laboratory_experiment; mixed	laboratory_1
STC-SINGLETON-SCR-00730	[110]	Continuous Event Recording Technology For Integrated Sensing And Communications In Space	FSO	shared_waveform; application_scenario	shared_experiment; analytical; simulation; field_experiment; mixed	field_1
STC-SINGLETON-SCR-00761	[111]	Joint Communication and Eavesdropper Detection on the Lossy Bosonic Channel	fiber	shared_waveform; shared_link_or_channel	shared_analytical; numerical_analysis	1
STC-SINGLETON-SCR-00808	[112]	Photonics-aided integrated sensing and communications in mmW bands based on a DC-offset QPSK-encoded LFM CW	photonic_THz	shared_waveform; shared_resource_allocation	shared_hardware; analytical; simulation; laboratory_experiment; prototype_testbed	1
STC-SINGLETON-SCR-00833	[113]	CORE-Lens: Simultaneous Communication and Object REcognition with Disentangled-GAN Cameras	VLC_LiFi	shared_waveform; shared_hardware	shared_dataset_based; laboratory_experiment; prototype_testbed	1
STC-SINGLETON-SCR-00836	[114]	Integrated Sensing and Communication in the Visible Spectral Range: A Novel Closed Loop Controller	VLC_LiFi	shared_hardware; shared_resource_allocation	shared_laboratory_experiment; prototype_testbed	1

Electronic Table ST-01A (continued): studies 99–105 of 206
Each study row cites its primary report and every eligible companion report; companions remain within the same study.

Study ID	Report citations	Title	Modality	Integration mechanisms	Validation	Reports
STC-SINGLETON-SCR-00873	[115]	Monolithically Integrated Sensing, Communication, and Energy Harvester	VLC_LiFi	shared_hardware	laboratory_experiment; prototype_testbed	1
STC-SINGLETON-SCR-00882	[116]	Performance Analysis of Microwave Photonics Assisted Joint Radar Communications	photonic_THz	shared_waveform; shared_resource_allocation	analytical; simulation	1
STC-SINGLETON-SCR-00884	[117]	Photonic-assisted generation of joint radar and communication signals with immunity to power fading	photonic_THz	shared_optical_carrier; shared_link_or_channel; shared_resource_allocation	analytical; simulation; laboratory_experiment	1
STC-SINGLETON-SCR-00885	[118]	Photonic-aid dual-formats LFM signals generator for joint radar-communication system	photonic_THz	shared_waveform; shared_hardware; shared_resource_allocation; joint_design_or_optimization	analytical; simulation	1
STC-SINGLETON-SCR-00887	[119]	Microwave Photonic Joint Radar, Wireless Communications, and Spectrum Sensing System With Broadband Tunability From 12 to 40 GHz	photonic_THz	shared_waveform; shared_hardware; shared_resource_allocation; joint_design_or_optimization	simulation; laboratory_experiment; mixed	1
STC-SINGLETON-SCR-00891	[120]	Millimeter-wave joint radar and communication system based on photonic frequency-multiplying constant envelope LFM-OFDM	photonic_THz	shared_waveform; shared_hardware; shared_resource_allocation; joint_design_or_optimization	simulation; laboratory_experiment; mixed	1
STC-SINGLETON-SCR-00893	[121]	Microwave Photonic Joint Radar and Secure Communication via Radar Signal Masking	photonic_THz	shared_hardware; shared_resource_allocation	laboratory_experiment; prototype_testbed	1

Electronic Table ST-01A (continued): studies 106–112 of 206
Each study row cites its primary report and every eligible companion report; companions remain within the same study.

Study ID	Report citations	Title	Modality	Integration mechanisms	Validation	Reports
STC-SINGLETON-SCR-00896	[122]	Photonics-assisted joint radar and communication system based on an optoelectronic oscillator	photonic_THz	shared_waveform; shared_resource_allocation	analytical; experiment; prototype_testbed	1
STC-SINGLETON-SCR-00899	[123]	Photonic-assisted generation of fast switchable multi-format LFM waveforms with a tuned bandwidth for a joint radar-communication system	photonic_THz	shared_waveform	analytical; simulation; prototype_testbed	1
STC-SINGLETON-SCR-00902	[124]	Bidirectional Retroreflective Optical ISAC Using Time Division Duplexing and Clipped OFDM	VLC_LiFi	shared_waveform; shared_link_or_channel; shared_resource_allocation	analytical; experiment	1
STC-SINGLETON-SCR-00903	[125]	Phase-Coded FMCW for Optical Integrated Sensing and Communication	FSO	shared_waveform; shared_hardware	analytical; simulation	1
STC-SINGLETON-SCR-00904	[126]	Gaussian Pulse-Based PSS-PPM Scheme for Optical Integrated Sensing and Communication Systems	FSO	shared_waveform	simulation	1
STC-SINGLETON-SCR-00905	[127]	Retroreflective Optical ISAC Supporting 3D Positioning in Indoor Environments	VLC_LiFi	shared_waveform; shared_hardware; shared_resource_allocation	analytical; simulation	1
STC-SINGLETON-SCR-00906	[128]	Quantum Optical Integrated Sensing and Communication With Homodyne BPSK Detection	other_optical	shared_waveform; joint_design_or_optimization	analytical; simulation	1

Electronic Table ST-01A (continued): studies 113–119 of 206
Each study row cites its primary report and every eligible companion report; companions remain within the same study.

Study ID	Report citations	Title	Modality	Integration mechanisms	Validation	Reports
STC-SINGLETON-SCR-00908	[129]	Optical ISAC: Fundamental Performance Limits and Transceiver Design	FSO	shared_waveform; shared_hardware; joint_design_or_optimization; shared_application_scenario	analytical; simulation	1
STC-SINGLETON-SCR-00909	[130]	Free Space Optical Integrated Sensing and Communication Based on LFM and CPM	FSO	shared_waveform; shared_hardware; joint_design_or_optimization	analytical; simulation	1
STC-SINGLETON-SCR-00910	[131]	Optical Integrated Sensing and Communication for Cooperative Mobile Robotics: Design and Experiments	VLC_LiFi	shared_hardware; shared_link_or_channel; shared_application_scenario	analytical; field_experiment	1
STC-SINGLETON-SCR-00912	[132]	Flexible Waveform Design for RO-ISAC Based on OFDM-Embedded Maximum Length Sequence	VLC_LiFi	shared_waveform; shared_resource_allocation	analytical; laboratory_experiment	1
STC-SINGLETON-SCR-00913	[133]	Pulse Sequence Sensing and Pulse Position Modulation for Optical Integrated Sensing and Communication	FSO	shared_waveform	analytical; simulation	1
STC-SINGLETON-SCR-00914	[134]	Optical Integrated Sensing and Communication System Based on Combination of OIRS and PD Array for Mobile Scenarios	FSO	shared_waveform; shared_resource_allocation	analytical; simulation	1
STC-SINGLETON-SCR-00915	[135]	UAV-Aided Flying Retroreflective Optical ISAC with Angle Diversity Transmitters	VLC_LiFi	shared_link_or_channel; shared_resource_allocation	simulation	1

Electronic Table ST-01A (continued): studies 120–126 of 206
Each study row cites its primary report and every eligible companion report; companions remain within the same study.

Study ID	Report citations	Title	Modality	Integration mechanisms	Validation	Reports
STC-SINGLETON-SCR-00916	[136]	Optical Integrated Sensing and Communication Lidar Based on RBPSK-FMCW Joint Waveform and Optical Phased Array	FSO	shared_waveform; shared_hardware; joint_design_or_optimization	analytical; simulation; laboratory_experiment	1
STC-SINGLETON-SCR-00917	[137]	Integrated Sensing and Communication for UAV Trajectory Optimization in Mixed FSO-RF Networks in Dynamic Weather Conditions	FSO	shared_resource_allocation; mixed	analytical; simulation; dataset_based	1
STC-SINGLETON-SCR-00918	[138]	Multiuser Retroreflective Optical ISAC: Waveform Design, Joint Sensing and Multiple Access	VLC_LiFi	shared_waveform; shared_resource_allocation	analytical; laboratory_experiment	1
STC-SINGLETON-SCR-00919	[139]	Full-Duplex RO-ISAC System: Wavelength Division Duplexing and Hybrid Waveform Design	VLC_LiFi	shared_resource_allocation	analytical; laboratory_experiment	1
STC-SINGLETON-SCR-00920	[140]	Experimental Demonstration of Multi-User Full-Duplex RO-ISAC System	VLC_LiFi	shared_waveform; shared_hardware; shared_resource_allocation	laboratory_experiment	1
STC-SINGLETON-SCR-00922	[141]	Leveraging ISAC for Adaptive Cell Zooming in Mixed FSO-RF Networks	FSO	shared_application_scenario; mixed	analytical; simulation; dataset_based	1
STC-SINGLETON-SCR-00925	[142]	Performance Analysis of NOMA-Assisted Optical OFDM ISAC Systems with Clipping Distortion	VLC_LiFi	shared_waveform; shared_resource_allocation	analytical; simulation	1

Electronic Table ST-01A (continued): studies 127–133 of 206
Each study row cites its primary report and every eligible companion report; companions remain within the same study.

Study ID	Report citations	Title	Modality	Integration mechanisms	Validation	Reports
STC-SINGLETON-SCR-00926	[143]	LACO-OFDM-Based Optical ISAC System Design and Performance Analysis	FSO	shared_resource_allocation; application_scenario	shared_analytical; simulation	1
STC-SINGLETON-SCR-00937	[144]	Simulation of Constant-Envelope THz Integrated Sensing and Communication System Based on Photonics with 2D Positioning	photonic_THz	shared_waveform; shared_resource_allocation; joint_design_or_optimization	analytical; simulation	1
STC-SINGLETON-SCR-00941	[145]	Joint Design of Optical Phased Array Beamforming and OFDM Waveform in Optical Wireless Integrated Sensing and Communication	FSO	shared_waveform; shared_optical_carrier; shared_resource_allocation; joint_design_or_optimization	shared_analytical; simulation	1
STC-SINGLETON-SCR-00943	[146]	Monolithic III-Nitride Photonic Circuit for Integrated Sensing and Communication System	VLC_LiFi	shared_hardware; shared_link_or_channel	laboratory_experiment; prototype_testbed	1
STC-SINGLETON-SCR-00944	[147]	Heterogeneous Integrated III-V-on-Soi Transmitter for 6G FiWi mmWave/FSO Integrated Sensing and Communication	hybrid_optical	shared_hardware; shared_link_or_channel; shared_resource_allocation	analytical; laboratory_experiment; prototype_testbed	1
STC-SINGLETON-SCR-00949	[148]	Double-effect DNN-based DBP Scheme for Integrated Sensing and Communications (ISAC)	fiber	shared_waveform	analytical; simulation	1
STC-SINGLETON-SCR-00953	[149]	A Beam-Scannable Photonic THz-ISAC System Based on Risley Prisms	photonic_THz	shared_waveform; joint_design_or_optimization	analytical; laboratory_experiment	1

Electronic Table ST-01A (continued): studies 134–140 of 206
Each study row cites its primary report and every eligible companion report; companions remain within the same study.

Study ID	Report citations	Title	Modality	Integration mechanisms	Validation	Reports
STC-SINGLETON-SCR-00954	[150]	Signal-Signal Beating Interference: From Destructive to Constructive for Photonic THz Integrated Sensing and Communication System Using Self-Coherent OFDM	photonic_THz	shared_resource_allocation	analytical; simulation; laboratory_experiment	1
STC-SINGLETON-SCR-00959	[151]	Fiber Events Recognition in Integrated Sensing and Communication Systems Based on Signal State of Polarization Analysis	fiber	shared_waveform	dataset_based; laboratory_experiment	1
STC-SINGLETON-SCR-00965	[152]	Microwave Photonics Empowered Minimalist Base Station for Integrated Sensing and Communication via Cascaded Intensity and Phase Modulation	photonic_THz	shared_optical_carrier; shared_resource_allocation	analytical; laboratory_experiment; prototype_testbed	1
STC-SINGLETON-SCR-00970	[153]	Spectral-Efficient Frequency-Division Photonic Millimeter-Wave Integrated Sensing and Communication System Using Improved Sparse LFM Sub-Bands Fusion	photonic_THz	shared_hardware; shared_resource_allocation; joint_design_or_optimization	analytical; laboratory_experiment; prototype_testbed	1
STC-SINGLETON-SCR-00974	[154]	Bit Inversion Probabilistic Shaping Enabling Performance Tradeoff in Photonics-Aided Millimeter-Wave OFDM Integrated Sensing and Communication Systems	photonic_THz	shared_waveform; shared_hardware; shared_resource_allocation	analytical; simulation; laboratory_experiment; prototype_testbed	1
STC-SINGLETON-SCR-00975	[155]	Radar-Centric Photonic Terahertz Integrated Sensing and Communication System Based on LFM-PSK Waveform	photonic_THz	shared_waveform; shared_hardware; joint_design_or_optimization	analytical; simulation; laboratory_experiment; prototype_testbed	1
STC-SINGLETON-SCR-00977	[156]	Photonics Terahertz Dual-Band Anti-Jamming Integrated Sensing and Communication System Empowered by Permutation Frequency Hopping	photonic_THz	shared_resource_allocation; joint_design_or_optimization	analytical; simulation; laboratory_experiment	1

Electronic Table ST-01A (continued): studies 141–147 of 206
Each study row cites its primary report and every eligible companion report; companions remain within the same study.

Study ID	Report citations	Title	Modality	Integration mechanisms	Validation	Reports
STC-SINGLETON-SCR-00979	[157]	Demonstration of Integrated Sensing and Communications in an ARoF-Based Radar System Utilizing Pulse Position Modulation	photonic_THz	shared_waveform	analytical; laboratory_experiment; field_experiment	1
STC-SINGLETON-SCR-00980	[158]	Interference Cancellation for Optical Wireless ISAC Based on Optical Phased Array	FSO	shared_waveform; joint_design_or_optimization	analytical; simulation	1
STC-SINGLETON-SCR-00983	[159]	A Spectrum-Efficient MoF Architecture for Joint Sensing and Communication in B5G Based on Polarization Interleaving and Polarization-Insensitive Filtering	fiber	shared_hardware; shared_resource_allocation	analytical; laboratory_experiment; prototype_testbed	1
STC-SINGLETON-SCR-00985	[160]	Photonic-Based Flexible Integrated Sensing and Communication With Multiple Targets Detection Capability for W-Band Fiber-Wireless Network	photonic_THz	shared_waveform; shared_link_or_channel; shared_resource_allocation	analytical; simulation; laboratory_experiment; prototype_testbed	1
STC-SINGLETON-SCR-00986	[161]	THz Integrated Sensing and Communication With Full-Photonic Direct LFM Reception and De-Chirping for D-Band Fiber-Wireless Network	photonic_THz	shared_link_or_channel; resource_allocation	analytical; simulation; laboratory_experiment; prototype_testbed	1
STC-SINGLETON-SCR-00990	[162]	Preamble-Free Synchronization Based on Dual-chirp Waveforms for Photonic THz-ISAC	photonic_THz	shared_waveform	analytical; simulation; laboratory_experiment; prototype_testbed	1
STC-SINGLETON-SCR-00992	[163]	Photonic Frequency-Hopped THz ISAC-Enabled by Electro-Optically Tunable Laser	photonic_THz	shared_hardware	analytical; laboratory_experiment; prototype_testbed	1

Electronic Table ST-01A (continued): studies 148–154 of 206
Each study row cites its primary report and every eligible companion report; companions remain within the same study.

Study ID	Report citations	Title	Modality	Integration mechanisms	Validation	Reports
STC-SINGLETON-SCR-00994	[164]	A Cross-Correlation MUSIC Scheme Based on OFDM for Photonics-Aided W-Band ISAC Systems	photonic_THz	shared_link_or_channel; resource_allocation	shared_analytical; experiment	laboratory_1
STC-SINGLETON-SCR-00996	[165]	Microwave Photonic Integrated Sensing and Communication Based on Polarization Multiplexing and Frequency-to-Time Mapping	photonic_THz	shared_resource_allocation	analytical; experiment	laboratory_1
STC-SINGLETON-SCR-00998	[166]	An Orthogonal Time Frequency Space Waveform With W-MUSIC Scheme for Photonic-Aided ISAC System	photonic_THz	shared_link_or_channel	analytical; experiment	laboratory_1
STC-SINGLETON-SCR-01000	[167]	Simultaneous Full-Photonic Dechirping and Demodulation THz-ISAC System Based on Thin-Film Lithium Niobate Modulator	photonic_THz	shared_link_or_channel	analytical; experiment; prototype_testbed	laboratory_1
STC-SINGLETON-SCR-01001	[168]	Movable Access Point-Aided Integrated Visible Light Communication and Sensing Networks	VLC_LiFi	shared_hardware; joint_design_or_optimization; shared_application_scenario	analytical; numerical_analysis; simulation	1
STC-SINGLETON-SCR-01004	[169]	Sliding Window-Based LSTM Scheme for PAM-4 Photonics-Assisted W-Band ISAC Systems	photonic_THz	shared_waveform; shared_link_or_channel	analytical; experiment; prototype_testbed	laboratory_1
STC-SINGLETON-SCR-01006	[170]	Co-Route Fiber Recognition and Status Diagnosis Based on Integrated Sensing and Communication in 6G Transport Networks	fiber	shared_link_or_channel; resource_allocation	analytical; dataset_based; field_experiment	1

Electronic Table ST-01A (continued): studies 155–161 of 206
Each study row cites its primary report and every eligible companion report; companions remain within the same study.

Study ID	Report citations	Title	Modality	Integration mechanisms	Validation	Reports
STC-SINGLETON-SCR-01008	[171]	Submarine integrated sensing and communication cable networks enable high-precision earthquake location estimation through linear wavefront theory	fiber	shared_link_or_channel	analytical; dataset_based; field_experiment	1
STC-SINGLETON-SCR-01009	[172]	Demonstration of Photonic Sub-THz ISAC System with Real-Time 251.03-Gbps Communication Rate and Offline 2.5-cm Sensing Resolution	photonic_THz	shared_link_or_channel; resource_allocation	shared_laboratory_experiment; prototype_testbed	1
STC-SINGLETON-SCR-01010	[173]	Photonic Chirp Spread Spectrum THz-ISAC: Constraints and Design Considerations	photonic_THz	shared_waveform; shared_resource_allocation	analytical; simulation; laboratory_experiment; prototype_testbed	1
STC-SINGLETON-SCR-01011	[174]	A THz with VLC Assisted ISAC System Under the Impact of Human Blockages	hybrid_optical	shared_application_scenario	analytical; simulation	1
STC-SINGLETON-SCR-01012	[175]	A Sub-THz ISAC System with Simultaneous Real-Time 125.516-Gbps Communication Rate and Offline 10-mm Sensing Resolution Enabled by Photonics	photonic_THz	shared_resource_allocation	analytical; laboratory_experiment	1
STC-SINGLETON-SCR-01018	[176]	Zero-Padding 2D-DFT Scheme for Photonics-Assisted W-Band Fiber-Wireless ISAC System With Orthogonal Time Frequency Space Modulation	photonic_THz	shared_link_or_channel	analytical; laboratory_experiment	1
STC-SINGLETON-SCR-01020	[177]	Probabilistic Shaping With Bit-Weighted Distribution Matching for Photonics-Assisted mmWave Dual-Functional OFDM ISAC Systems	photonic_THz	shared_waveform; shared_hardware; shared_resource_allocation	analytical; simulation; laboratory_experiment	1

Electronic Table ST-01A (continued): studies 162–168 of 206
Each study row cites its primary report and every eligible companion report; companions remain within the same study.

Study ID	Report citations	Title	Modality	Integration mechanisms	Validation	Reports
STC-SINGLETON-SCR-01021	[178]	Microalgae Density Measurement Using Quantum-Dot-Integrated Sensing and Communication System	VLC_LiFi	shared_waveform	analytical; simulation; laboratory_experiment	1
STC-SINGLETON-SCR-01023	[179]	Passive Sensing Technique Using Cyclic Prefix in Power-Constrained Optical Wireless Communication Systems	FSO	shared_waveform; shared_resource_allocation	analytical; simulation	1
STC-SINGLETON-SCR-01024	[180]	Single-Source Integrated Optical Wireless Positioning and Communication	VLC_LiFi	shared_waveform; shared_resource_allocation	analytical; simulation	1
STC-SINGLETON-SCR-01025	[181]	Revolutionizing 6G: Experimental Validation of an Optical Integrated Communication, Sensing, and Power Transfer System	FSO	shared_waveform; shared_resource_allocation	analytical; simulation; laboratory_experiment; prototype_testbed	1
STC-SINGLETON-SCR-01028	[182]	Adaptive Lighting Control in Visible Light Systems: An Integrated Sensing, Communication, and Illumination Framework	VLC_LiFi	shared_resource_allocation	analytical; numerical_analysis; simulation	1
STC-SINGLETON-SCR-01029	[183]	Experimental Demonstration of Integrated Optical Wireless Sensing and Communication	FSO	shared_waveform; shared_resource_allocation	laboratory_experiment; prototype_testbed	1
STC-SINGLETON-SCR-01030	[184]	Joint Optimization of Slot Selection and Power Allocation in Integrated Visible Light Communication and Sensing Systems	VLC_LiFi	shared_waveform; shared_resource_allocation	analytical; simulation	1

Electronic Table ST-01A (continued): studies 169–175 of 206
Each study row cites its primary report and every eligible companion report; companions remain within the same study.

Study ID	Report citations	Title	Modality	Integration mechanisms	Validation	Reports
STC-SINGLETON-SCR-01032	[185]	Flexible Hybrid Waveform Design for Optical Integrated Communication, Sensing, and Power Transfer Systems	FSO	shared_hardware; shared_link_or_channel; shared_resource_allocation	analytical; simulation; laboratory_experiment; prototype_testbed	1
STC-SINGLETON-SCR-01033	[186]	Integrated PC-FMCW Automotive Headlamp for Communication, Sensing, and Illumination	VLC_LiFi	shared_hardware; shared_application_scenario	analytical; simulation	1
STC-SINGLETON-SCR-01034	[187]	VehicleTalk: Lightweight V2V Network Enabled by Optical Wireless Communication and Sensing	VLC_LiFi	shared_waveform; shared_hardware; shared_application_scenario	analytical; laboratory_experiment; prototype_testbed	1
STC-SINGLETON-SCR-01036	[188]	ReflexGest: Recognizing Hand Gestures Under VLC-Capable Lamps	VLC_LiFi	shared_hardware	analytical; laboratory_experiment; prototype_testbed	1
STC-SINGLETON-SCR-01040	[189]	A Novel Comb Based Multi-Function Architecture for ISAC System	photonic_THz	shared_link_or_channel	analytical; simulation; laboratory_experiment	1
STC-SINGLETON-SCR-01041	[190]	Photonics-based Terahertz Integrated Communication and Localization for D-band	photonic_THz	shared_waveform; shared_hardware	laboratory_experiment	1
STC-SINGLETON-SCR-01045	[191]	Secure Multi-Tenant Distributed Sensing for ISAC-Over-Fiber (IOSAC) Networks	fiber	shared_resource_allocation; shared_application_scenario	simulation	1

Electronic Table ST-01A (continued): studies 176–182 of 206
Each study row cites its primary report and every eligible companion report; companions remain within the same study.

Study ID	Report citations	Title	Modality	Integration mechanisms	Validation	Reports
STC-SINGLETON-SCR-01047	[192]	Indoor Health Monitoring with VLC-based Passive Posture Monitoring	VLC_LiFi	shared_waveform	analytical; dataset_based; laboratory_experiment	1
STC-SINGLETON-SCR-01049	[193]	Optical Transport Networks Supporting Integrated Communications and Sensing in 6G	fiber	shared_waveform; shared_resource_allocation; shared_application_scenario	analytical; numerical_analysis; prototype_testbed	1
STC-SINGLETON-SCR-01050	[194]	Hybrid MRR-Assisted Long-Range SDMA for RO-ISAC Systems in Industry IoT Applications	FSO	shared_resource_allocation	simulation; prototype_testbed	1
STC-SINGLETON-SCR-01059	[195]	Joint Task Offloading and Resource Allocation for Integrated VLC and Sensing in Digital Twin-Aided Vehicular Edge Computing Networks	VLC_LiFi	shared_resource_allocation; joint_design_or_optimization; shared_application_scenario	analytical; simulation	1
STC-SINGLETON-SCR-01060	[196]	An Information Interaction Method for Autonomous Train Operation Protection Based on Detection and Communication Integrated Microwave Photonic Radar	photonic_THz	shared_hardware; shared_resource_allocation	analytical; simulation; laboratory_experiment	1
STC-SINGLETON-SCR-01070	[197]	Integrated 1-Tb/s/λ Coherent Optical Transmission and Forward Vibration Sensing Using DFB Lasers with Combined Linewidth of 21.4 MHz	fiber	shared_optical_carrier; shared_link_or_channel	analytical; laboratory_experiment	1
STC-SINGLETON-SCR-01071	[198]	Long-Distance Integrated Forward Vibration Sensing and Terabit Optical Communication with 12.5-MHz DFB Laser via Phase Noise Separation	fiber	shared_optical_carrier; shared_link_or_channel	analytical; laboratory_experiment	1

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Each study row cites its primary report and every eligible companion report; companions remain within the same study.

Study ID	Report citations	Title	Modality	Integration mechanisms	Validation	Reports
STC-SINGLETON-SCR-01072	[199]	Accurate vibration recognition enabled by synergistic sensing based on both phase and SOP for stable IM-DD optical interconnects	fiber	shared_waveform; optimization	dataset_based; experiment	laboratory_1
STC-SINGLETON-SCR-01073	[200]	Photonics-Assisted System for Joint Radar Ranging, Wireless Communications, and Spectrum Sensing	photonic_THz	shared_resource_allocation	laboratory_experiment	1
STC-SINGLETON-SCR-01079	[201]	Sensing as a Service: Can In-Band Distributed Fiber Optic Sensing Coexist With Coherent Metro DWDM Communication Network?	fiber	shared_link_or_channel; resource_allocation	simulation; experiment	laboratory_1
STC-SINGLETON-SCR-01081	[202]	Visible Light Integrated Positioning and Communication: A Multi-Task Federated Learning Framework	VLC_LiFi	shared_waveform; allocation; optimization	shared_resource_; joint_design_or_	analytical; simulation1
STC-SINGLETON-SCR-01085	[203]	Turbulence-Resilient Communication through Beam Shaping Enabled by Multi-Region Turbulence Intensity Sensing	FSO	shared_link_or_channel	laboratory_experiment	1
STC-SINGLETON-SCR-01086	[204]	Simultaneous distributed sensing and communication enabled by space and wavelength division multiplexing in a field-deployed seven-core fiber	fiber	shared_link_or_channel; resource_allocation	shared_analytical; experiment; field_experiment	laboratory_1
STC-SINGLETON-SCR-01088	[205]	Picosecond-Level Round-Trip Delay Measurement Scheme Based on Non-Commensurate Sampling for Future Deterministic Passive Optical Networks	fiber	shared_hardware	analytical; simulation; laboratory_experiment	1

Electronic Table ST-01A (continued): studies 190–196 of 206
Each study row cites its primary report and every eligible companion report; companions remain within the same study.

Study ID	Report citations	Title	Modality	Integration mechanisms	Validation	Reports
STC-SINGLETON-SCR-01093	[206]	Coexistence of Distributed Sensing and Ultra-Simple Coherent RoF Based Front-Haul for Radio Access Network	photonic_THz	shared_optical_carrier; shared_link_or_channel	analytical; experiment	laboratory_ 1
STC-SINGLETON-SCR-01095	[207]	Advanced Integration of Sensing and Communication With High DSP Compatibility for SMART Network	fiber	shared_resource_allocation	laboratory_experiment	1
STC-SINGLETON-SCR-01101	[208]	Integrated Communication and Radar Detection Based-on Single Modulator at W-Band	photonic_THz	shared_resource_allocation	simulation; experiment	laboratory_ 1
STC-SINGLETON-SCR-01103	[209]	Vibration Localization under Unidirectional Single-Channel Transmission Conditions in the DSCM Systems Based on Non-Reciprocity	fiber	shared_waveform; shared_resource_allocation	simulation	1
STC-SINGLETON-SCR-01105	[210]	Integrated secure communication and sensing based on simultaneous chaotic encryption and vibration detection with an RE-LFM signal	fiber	shared_link_or_channel; shared_resource_allocation	laboratory_experiment	1
STC-SINGLETON-SCR-01151	[211]	Photonic Millimeter-Wave Joint Radar Communication System Using Spectrum-Spreading Phase-Coding	photonic_THz	shared_waveform; shared_optical_carrier	shared_hardware; laboratory_experiment	1
STC-SINGLETON-SCR-01152	[212]	On-Chip Photonic MMW Joint Communication and Radar System	photonic_THz	shared_hardware; shared_resource_allocation	laboratory_experiment	1

Electronic Table ST-01A (continued): studies 197–203 of 206
Each study row cites its primary report and every eligible companion report; companions remain within the same study.

Study ID	Report citations	Title	Modality	Integration mechanisms		Validation	Reports
STC-SINGLETON-SCR-01153	[213]	Photonic-Assisted Joint Radar and Communication System Using Overlapped Spectrum for High-Resolution Ranging and High-Speed Data Transmission	photonic_THz	shared_hardware;	shared_resource_allocation	laboratory_experiment	1
STC-SINGLETON-SCR-01165	[214]	Joint radar and communication system based on a chaotic optoelectronic oscillator	photonic_THz	shared_waveform;	joint_design_or_optimization	laboratory_experiment	1
STC-SINGLETON-SCR-01180	[215]	Optical fiber communication and distributed multi-parameter sensing integrated network for smart power grid applications	fiber	shared_link_or_channel;	shared_resource_allocation;	analytical; laboratory_experiment; field_experiment	1
STC-SINGLETON-SCR-01188	[216]	Seamless integration of distributed acoustic sensing and passive optical networks for human intrusion monitoring	fiber	shared_resource_allocation		analytical; laboratory_experiment; field_experiment	1
STC-SINGLETON-SCR-01229	[217]	Photonic generation of frequency agile LFM signals for ISAC systems	hybrid_optical	shared_waveform;	shared_hardware;	analytical; simulation; laboratory_experiment; mixed	1
STC-SUBCARRIER-CHIRP-DBAN-D-2025	[218], [219]	Photonics-Aided THz Integrated Sensing and Communication System Based on a Subcarrier-Chirp Inter-Embedded Waveform	photonic_THz	shared_optical_carrier;	shared_resource_allocation	analytical; laboratory_experiment; prototype_testbed	2
STC-SUBMARINE-CABLE-DAS-HUIZHOU-2022-2026	[220]	Photonic Integrated Sensing and Communication System Harnessing Submarine Fiber Optic Cables for Coastal Event Monitoring	fiber	shared_link_or_channel;	shared_resource_allocation	dataset_based; field_experiment	1

Electronic Table ST-01A (continued): studies 204–206 of 206
Each study row cites its primary report and every eligible companion report; companions remain within the same study.

Study ID	Report citations	Title	Modality	Integration mechanisms	Validation	Reports
STC-TUNABLE-LASER-THZ-FMCW-2025-2026	[221], [222]	Terahertz ISAC with Simultaneous Fast-Swept FMCW Radar and High-Speed Wireless Link Using a Single UTC-PD	photonic_THz	shared_waveform; allocation; optimization	shared_resource_joint_design_or_ laboratory_experiment	2
STC-WBAND-POLARIZATION-JR-C-2023-2025	[223], [224]	Integration of Sensing and Communication in a W-Band Fiber-Wireless Link Enabled by Electromagnetic Polarization Multiplexing	photonic_THz	shared_hardware; allocation	shared_resource_ laboratory_experiment	2
STC-WBAND-TFDM-DONG-2022-2024	[225], [226], [227]	Photonic-Based W-Band Integrated Sensing and Communication System With Flexible Time- Frequency Division Multiplexed Waveforms for Fiber-Wireless Network	photonic_THz	shared_link_or_channel; resource_allocation	shared_analytical; simulation; laboratory_experiment; prototype_testbed	3

Electronic Table ST-01B: Exact provenance of the 21 eligible companion reports
E/M/T gives extracted evidence, metric, and governed tradeoff row counts. Guard identifies report-specific metric guardrails.

Study ID	Citation	Companion role	E/M/T	Guard	Pending	Synthesis use gate
STC-DP-IMDD-THZ-RANGING-2025-2026	[3]	conference_version	3/1/0	0	0	eligible_with_study_context
STC-DSCM-ANN-DT-ECL-2025-2026	[5]	conference_version	2/7/0	0	0	eligible_with_study_context
STC-EADO-OFDM-RANGING-2024-2025	[7]	conference_version	4/7/1	0	0	eligible_with_study_context
STC-ENDOGENOUS-TS-FRFT-DAS-2025	[9]	conference_companion_report	2/5/1	5	0	report_specific_guardrail
STC-FRDM-POLARIZATION-SENSING-2025-2026	[11]	conference_companion_report	2/4/1	2	0	report_specific_guardrail
STC-III-NITRIDE-MOTION-VLC-2023	[14]	companion_report	10/16/1	0	0	eligible_with_study_context
STC-MMW-OFDM-PENG-2023-2024	[17]	conference_version	4/12/1	0	0	eligible_with_study_context

Electronic Table ST-01B (continued): companions 8–14 of 21
E/M/T gives extracted evidence, metric, and governed tradeoff row counts. Guard identifies report-specific metric guardrails.

Study ID	Citation	Companion role	E/M/T	Guard	Pending	Synthesis use gate
STC-MULTICHANNEL-LFM-QA M-THZ-2024-2026	[19]	conference_version	2/1/0	0	0	eligible_with_study_context
STC-OCDM-THZ-OVER-FIBER-2 024-2025	[21]	conference_version	3/1/0	0	0	eligible_with_study_context
STC-ODDM-PHOTONIC-MMW-2 024-2025	[23]	conference_version	2/2/0	0	0	eligible_with_study_context
STC-OPA-OWISAC-BEAMFORM ING-2024-2025	[25]	conference_version	4/5/0	0	0	eligible_with_study_context
STC-REALTIME-BIDIR-DMT-TH Z-2024-2026	[27]	conference_version	4/1/0	0	0	eligible_with_study_context
STC-REALTIME-BIDIR-DMT-TH Z-2024-2026	[28]	companion_report	4/1/0	0	0	eligible_with_study_context
STC-SELF-HOMODYNE-MCF-OF DR-2024-2025	[30]	conference_version	4/2/0	0	0	eligible_with_study_context

Electronic Table ST-01B (continued): companions 15–21 of 21
E/M/T gives extracted evidence, metric, and governed tradeoff row counts. Guard identifies report-specific metric guardrails.

Study ID	Citation	Companion role	E/M/T	Guard	Pending	Synthesis use gate
STC-SEVENCORE-COMB-SH-DA S-2024-2025	[32]	conference_version	3/0/0	0	0	eligible_with_study_context
STC-SF-LFM-OFDM-THZ-2025-2 026	[34]	conference_version	5/14/1	0	0	eligible_with_study_context
STC-SUBCARRIER-CHIRP-DBA ND-2025	[219]	conference_version	4/8/0	0	0	eligible_with_study_context
STC-TUNABLE-LASER-THZ-FM CW-2025-2026	[222]	conference_companion_report	19/11/1	7	0	report_specific_guardrail
STC-WBAND-POLARIZATION-J RC-2023-2025	[224]	conference_version	2/4/0	0	0	eligible_with_study_context
STC-WBAND-TFDM-DONG-2022 -2024	[226]	conference_version	2/4/0	0	0	eligible_with_study_context
STC-WBAND-TFDM-DONG-2022 -2024	[227]	companion_report	10/17/2	0	0	eligible_with_study_context

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